AZURE DEVOPS, CI/CD AND AI-ASSISTED RELEASE READINESS — DAY 5

Module 8: AI-Assisted Release Readiness & Quality Intelligence

Turning pipeline, test and change signals into a risk score, a drafted changelog and a go/no-go scorecard — with a human holding the final decision.

MODULE 8 OF 8

RECAP

From automated pipelines to an informed go/no-go

Modules 1–7 built the pipeline, the quality gates and the AI-assisted dev workflow. Module 8 turns that signal into one decision-support tool for release day.

M1	M2	M3	M4	M5	M6	M7	M8
Foundations & Traceability	Branching & PR Gates	CI Pipelines (YAML)	Testing & Quality Gates	CD & Deployment	IaC & DevSecOps	AI-Assisted Dev	**Release Readiness — today**

LEARNING OBJECTIVES

By the end of this module, you can...

- Define what “release-ready” means and the signals that measure it
- Generate an AI-assisted change-risk score, with hotspot detection
- Auto-draft release notes and change summaries from the diff
- Use AI to triage failures and suggest likely root causes
- Spot anomalies in pipeline and quality trends before they bite
- Combine it all into a go/no-go scorecard — with a human sign-off

DEFINING THE TARGET

What “release-ready” actually means

Not just “tests pass”. Four signal families, measured together, are what a real go/no-go decision rests on.

Coverage & quality gates

Test results, coverage thresholds, static analysis — the gates from Module 4

Change risk

How big, how risky and how well-tested is this specific change set

Traceability

Every change linked back to a work item, a PR and an approved build

Operational history

DORA signals — deployment frequency, lead time, change-fail rate, MTTR

CHANGE RISK

Scoring risk before a change ships

Inputs

- Diff size & complexity
- Files touched vs. known hotspots
- Coverage delta on changed lines
- Author & reviewer history

AI scoring

Azure OpenAI reads the change set and prior incident history, and produces a weighted risk score with a plain-language rationale

Risk band

MEDIUM

RISK-SCORE.PS1

The real script behind that risk score

Inputs

- Completed PRs closed in the last 14 days
- Each PR's changed files, from its latest iteration
- Hotspot check — does the diff touch order_totals.py?
- No coverage delta or reviewer history fetched today

AI scoring

Azure OpenAI reads the per-PR file-change summary, rates risk LOW / MEDIUM / HIGH, and flags any PR touching the hotspot file

Risk band

MEDIUM

RELEASE NOTES

Let AI draft the changelog — you edit, not author

The same diff-summarisation pattern from Module 7, scaled up: every merged PR since the last release, rolled into one structured note.

INPUT

14 merged PRs — their titles, descriptions and diffs — since v2.3.0

v2.4.0 — DRAFT

Features

- Prescription refill reminders (PBI-482)

Fixes

- Negative-quantity orders now rejected (BUG-117)

Breaking changes

- None detected

RELEASE-NOTES.PS1

The real script behind that changelog

Fetches completed PRs closed since --Since (default 14 days) and asks Azure OpenAI to group them into Features, Fixes and Breaking changes.

INPUT

Completed PRs since --Since: title, description, closedDate

v1.1.0 — DRAFT

Features

- One line per merged PR — plain and specific

Fixes

- Plain ASCII punctuation only — no em dashes, no curly quotes

Output file

- Saved to release-notes-draft.md

FAILURE TRIAGE

When a build breaks, get to “why” faster

- A build or test stage fails in the pipeline
- AI reads the log and stack trace, and clusters it against known failure signatures
- It suggests a likely root cause and links similar past failures — as a hint, not a fix
- **An engineer confirms the cause and decides the fix — AI never auto-commits a change**

ANOMALY DETECTION

Spotting drift before it becomes an incident

A threshold catches what's already broken. Anomaly detection catches what's quietly getting worse.

- Build duration creeping up week over week
- Flaky-test rate rising on one service
- Coverage trending down on a critical module
- Deployment frequency or lead time (DORA) drifting from baseline

THE SCORECARD

Four inputs, one recommendation

- Coverage & gates — pass
- Change risk — medium
- Traceability — complete
- Pipeline health — stable



READINESS SCORE

82

GO — WITH CONDITIONS

SCORECARD.PS1

The real script behind the scorecard

- Risk report — risk-score.md, if present
- Release notes — release-notes-draft.md, if present
- Traceability — linked work items on up to 50 PRs
- Build health — last 5 builds, status & result



READINESS SCORE

82

GO — WITH CONDITIONS

THE HONEST GAP

These scripts don't gate the pipeline — yet

What the scripts do

- Write risk-score.md, release-notes-draft.md, scorecard.md
- Print the same report to the console
- Read Script 1/2's output, if present, for the scorecard
- Ask AI for a Score: and Recommendation: line

What they don't do

- Check whether the recommendation is GO or NO-GO
- Set a pipeline result or return a failure code
- Stop the pipeline passing on a NO-GO scorecard
- Fail for anything but an actual script error

CLOSING THE GAP

Add one pipeline step to make it a real gate

A final PowerShell step reads scorecard.md and fails the build on NO-GO. GO - WITH CONDITIONS still passes — add an environment approval there for a human sign-off.

THE STEP

A PowerShell@2 task, added as the last step after all three scripts run

PowerShell@2 — inline script

Reads

- $(Build.SourcesDirectory)\scorecard.md

Matches

- Recommendation: GO | GO - WITH CONDITIONS | NO-GO

Throws the build on

- NO-GO, or no recognized line at all (fail closed)

GUARDRAILS

AI recommends. A person decides.

What AI does

- Scores risk and flags hotspots
- Drafts release notes
- Suggests root causes
- Flags anomalies against baseline

What the human owns

- Reviews and edits every draft
- Confirms the actual root cause
- Weighs risk against context AI can't see
- Makes — and is accountable for — the go/no-go call

CASE STUDY

Build the AI-assisted release-readiness check

01

Score the release

An AI risk summary over the full change set and test signals since the last release

02

Auto-draft the release notes

A structured changelog generated from merged PRs, ready for a human edit pass

03

Surface the scorecard

Go / no-go, with its evidence, presented for human approval before release

PRACTICE LAB

What we'll use

- Azure OpenAI — risk summaries, release notes, triage hints
- Azure DevOps REST APIs — pulling PRs, builds, tests, work items
- DORA & Analytics — deploy frequency, lead time, change-fail rate, MTTR
- The release-readiness scorecard — the artifact we ship

SHARED SETUP

Shared setup and run order

- ADO_PAT — Azure DevOps auth, used by all three scripts
- AOAI_ENDPOINT & AOAI_KEY — same Azure OpenAI resource as Module 7
- Run order: risk-score.ps1 → release-notes.ps1 → scorecard.ps1
- Still runs without them — just with less signal

KEY TAKEAWAYS

From signal to decision

- Release-ready is measurable, not a feeling
- AI concentrates the evidence — it doesn't make the call
- This scorecard is the backbone of tomorrow's Capstone

NEXT: CAPSTONE